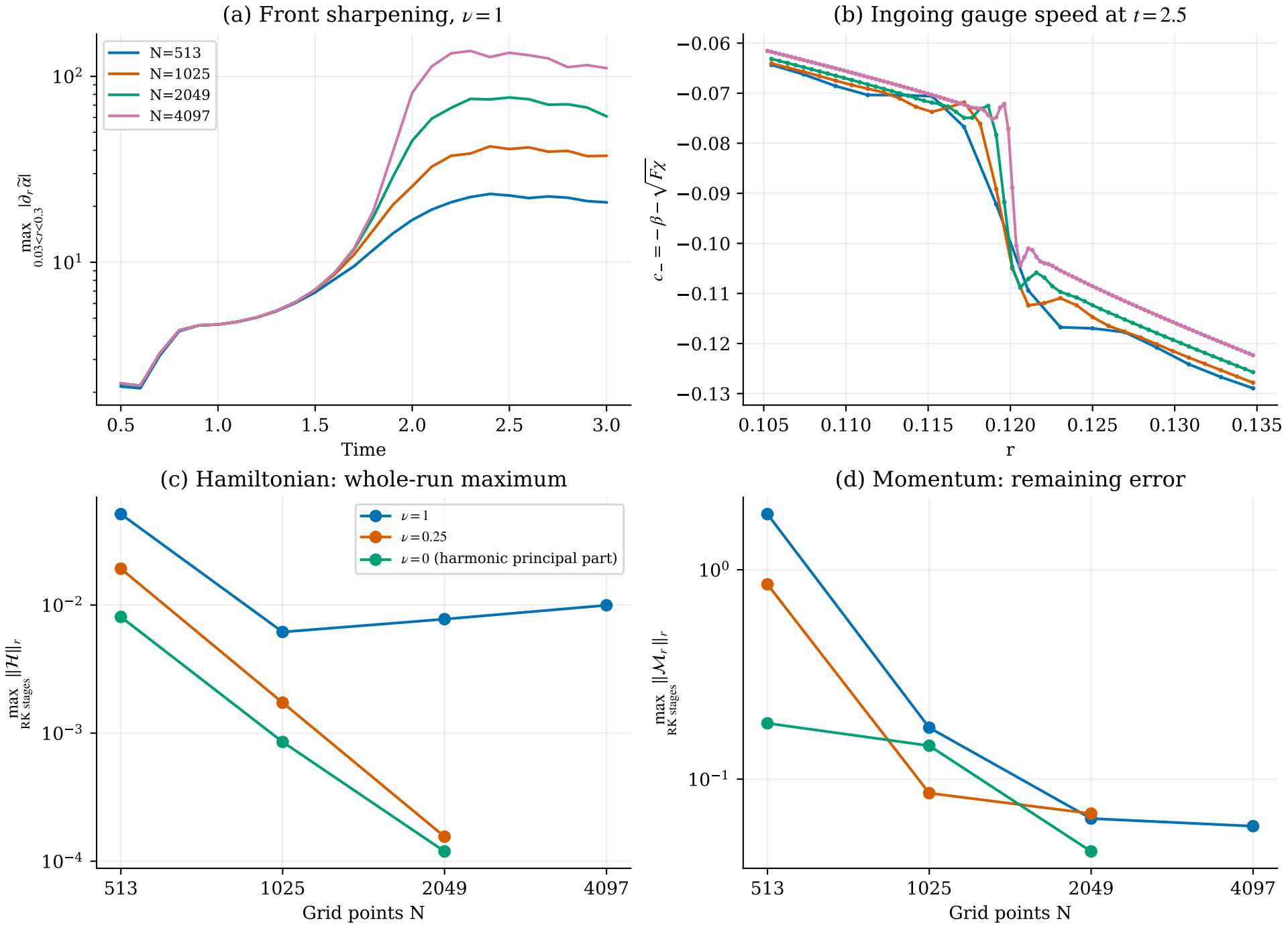
(a) Front sharpening, $\nu = 1$
N=513
N=1025
N=2049
N=4097
10^2
10^1
$\max_{0.03<r<0.3} |\partial_r \tilde{\alpha}|$
0.5
1.0
1.5
2.0
2.5
3.0
Time
(b) Ingoing gauge speed at $t = 2.5$
$c_- = -\beta - \sqrt{F\chi}$
−0.06
−0.07
−0.08
−0.09
−0.10
−0.11
−0.12
−0.13
0.105
0.110
0.115
0.120
0.125
0.130
0.135
r
(c) Hamiltonian: whole-run maximum
$\nu = 1$
$\nu = 0.25$
$\nu = 0$ (harmonic principal part)
10^{-2}
10^{-3}
10^{-4}
$\max_{\text{RK stages}} \|\mathcal{H}\|_r$
513
1025
2049
4097
Grid points N
(d) Momentum: remaining error
10^0
10^{-1}
$\max_{\text{RK stages}} \|\mathcal{M}_r\|_r$
513
1025
2049
4097
Grid points N